



Predicting Monthly Airline Departures Using Machine Learning

Applied Data Science

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BUSINESS PROBLEM

ACCURATELY FORECASTING ROUTE DEMAND IS ESSENTIAL FOR PROFITABLE AIRLINE SCHEDULING DECISIONS



Airlines must decide:

-  Which routes should be served?
-  How often should flights operate?
-  How can schedules maximize profitability?

Business Value:

- Improve route planning
- Support fleet allocation
- Enhance operational efficiency



DATASET & DATA PREPARATION

MORE THAN 250,000 AIRLINE SCHEDULING RECORDS WERE PREPARED FOR MACHINE LEARNING ANALYSIS

Table 1. Dataset Summary

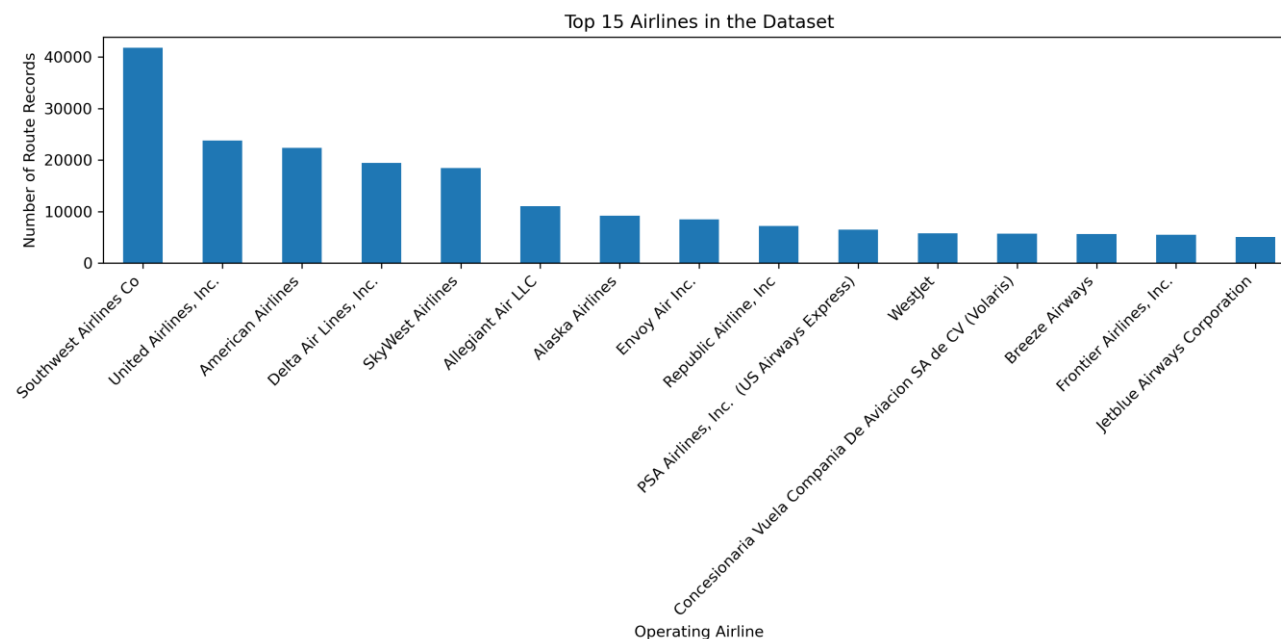
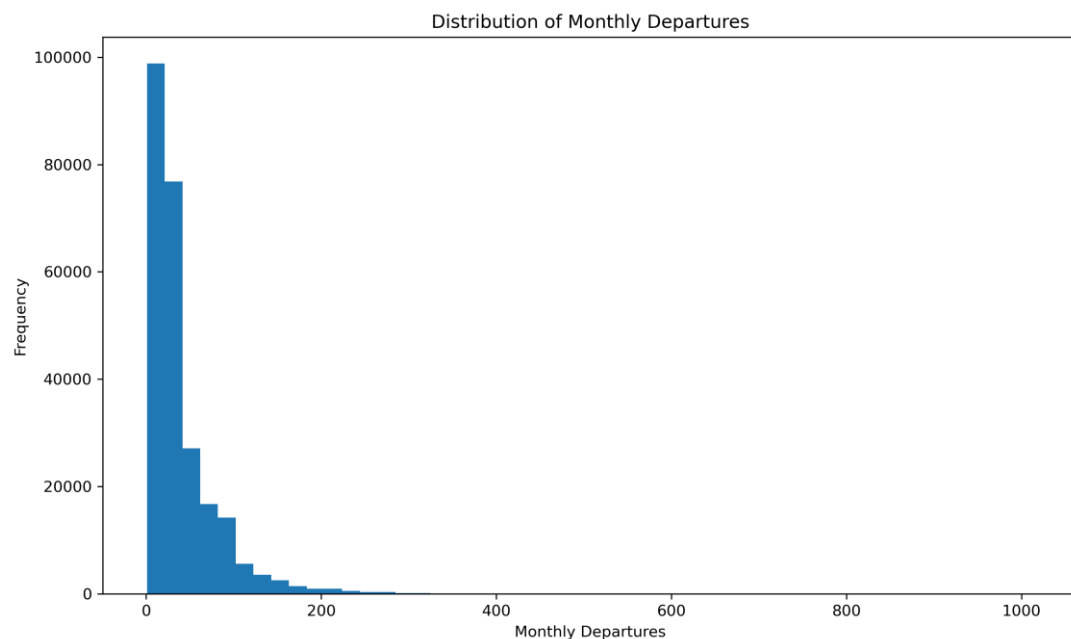
Data Preparation:

- 1 Removed missing values
- 2 Removed redundant variables
- 3 Encoded categorical variables

Metric	Value
Original Records	250786
Original Variables	24
Final Modeling Records	250786
Final Modeling Variables	20
Training Records	200628
Testing Records	50158
Missing Values Removed	232



EXPLORATORY DATA ANALYSIS



Key Findings:

- ✓ Monthly departures were highly right-skewed
- ✓ Major airlines dominated scheduling activity
- ✓ The dataset represents a diverse mix of commercial airlines
- ✓ Operational patterns supported predictive modeling



MACHINE LEARNING MODELS

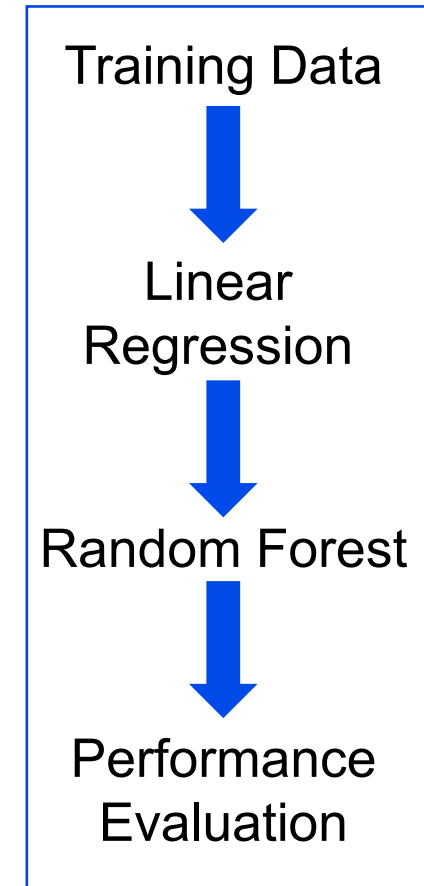
BOTH MODELS WERE TRAINED AND EVALUATED USING THE SAME DATA AND PERFORMANCE METRICS TO ENSURE A FAIR COMPARISON

Linear Regression	Random Forest
Baseline Model	Ensemble Model
Models Linear Relationship	Models Nonlinear Relationship
Easy to Interpret	Captures Complex Interactions

Evaluation Metrics:

- ✓ Mean Absolute Error (MAE)
- ✓ Root Mean Squared Error (RMSE)
- ✓ Coefficient of Determination (R^2)

Development Workflow:



UNEXPECTED DISCOVERY

WERE THESE RESULTS TOO GOOD TO BE TRUE?

Table 3. Initial Random Forest Performance

Metric	Value
Mean Absolute Error (MAE)	0.05
Mean Squared Error (MSE)	0.37
Root Mean Squared Error (RMSE)	0.61
R ²	0.9998

The Investigation:

Reviewed Feature
Importance



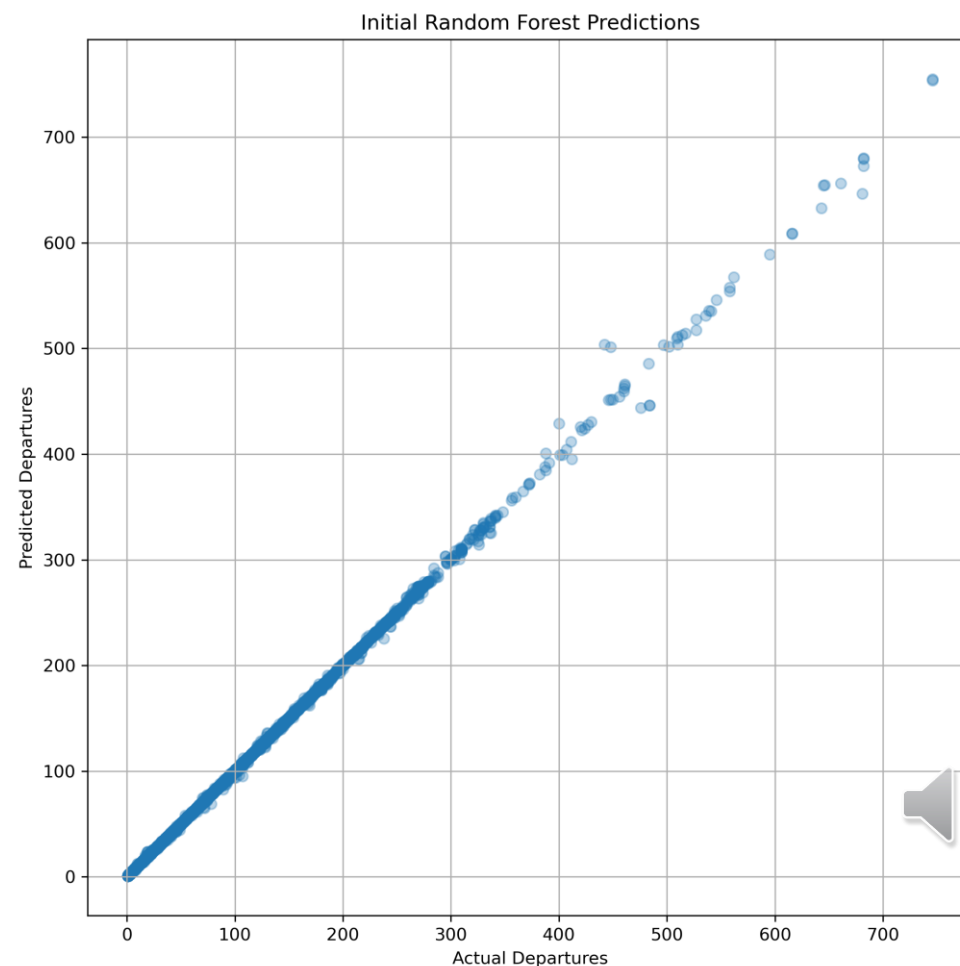
**Identified Data
Leakage**



Removed Leakage
Variable



Retrained the
model



FINAL MODEL RESULTS

REMOVING DATA LEAKAGE PRODUCED A REALISTIC AND RELIABLE MODEL
SUITABLE FOR DECISION SUPPORT

Table 5. Model Comparison

Model	MAE	RMSE	R ²	Notes
Linear Regression	8.97	16.78	0.8545	Baseline model
Initial Random Forest	0.05	0.61	0.9998	Inflated by data leakage
Revised Random Forest	9.71	16.01	0.8675	Final selected model

Most Important Predictors:

Route Distance

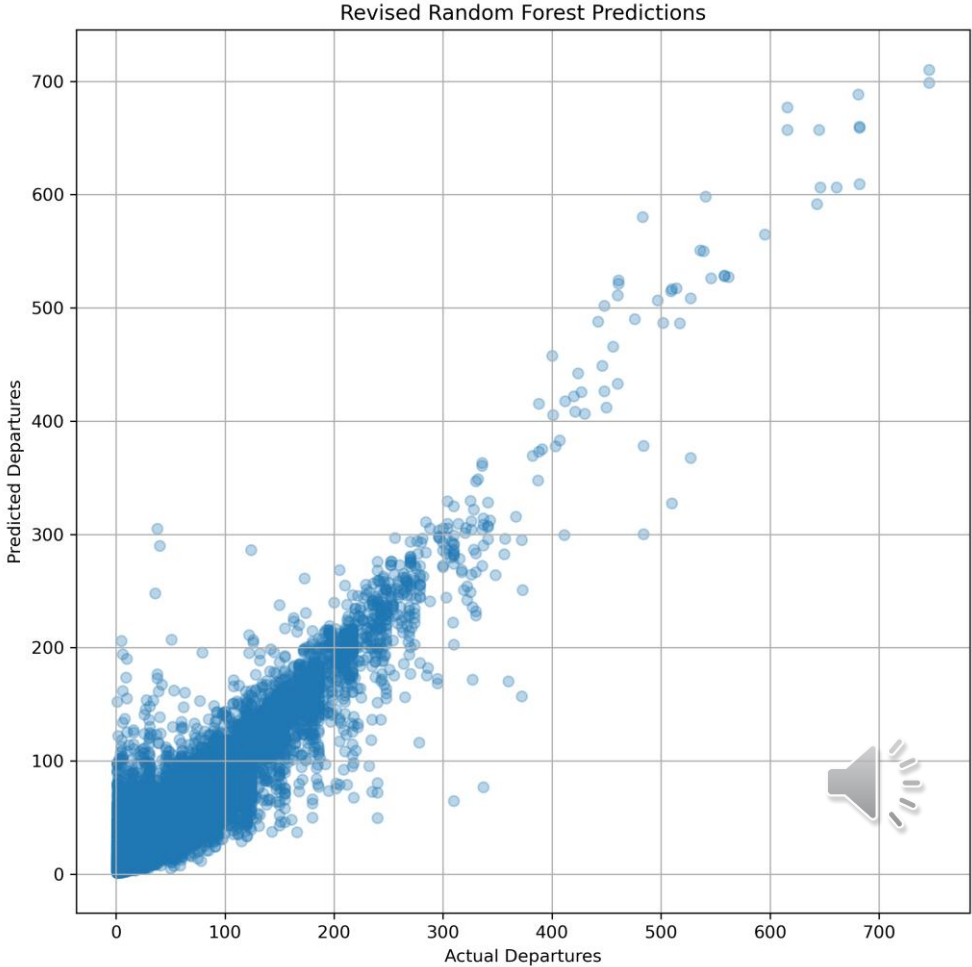
Operating Airline

Origin Airport

Destination Airport

Seats per Operation

Months



BUSINESS RECOMMENDATIONS

THE VALIDATED MODEL SUPPORTS DATA-DRIVEN AIRLINE SCHEDULING AND NETWORK PLANNING

Airlines Can Use this Model To:

- ✓ Improve network planning
- ✓ Optimize fleet allocation
- ✓ Support schedule planning
- ✓ Forecast route demand
- ✓ Improve resource utilization

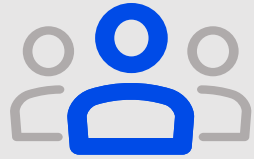
The model should be used as a decision-support tool to complement, not replace, airline planning expertise

- 1 Update Schedule Data
- 2 Retrain the Model
- 3 Validate Performance
- 4 Deploy as Decision Support
- 5 Monitor, Retrain & Improve



ETHICAL CONSIDERATIONS

RESPONSIBLE MACHINE LEARNING REQUIRES TRANSPARENCY, FAIRNESS, HUMAN OVERSIGHT, AND CONTINUOUS EVALUATION



Human Oversight

Machine learning should support, not replace, experienced airline planners



Transparent Validation

Model validation is equally important as predictive accuracy



Fair and Unbiased Analysis

Identifying and removing data leakage ensured fair, reliable, and trustworthy predictions



Continuous Model Monitoring

Models should be continuously monitored and retrained as airline operations and market conditions evolve



CONCLUSION

THIS PROJECT DEMONSTRATES HOW VALIDATED MACHINE LEARNING CAN SUPPORT SMARTER, DATA-DRIVEN AIRLINE SCHEDULING DECISIONS

✓ **Machine learning
can accurately
predict monthly
airline departures**

✓ **The revised
Random Forest
model delivered
the strongest
predictive
performance after
removing data
leakage**

✓ **Rigorous model
validation
produced reliable
business insights**

THANK YOU!

